

Bayesian Normal-Normal Model

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Observations: $y = \{y_1, y_2, \dots, y_n\}$ from a Normal population $N(\mu, \sigma^2)$.

$$y_i \mid \mu, \sigma^2 \stackrel{iid}{\sim} N(\mu, \sigma^2)$$
$$\mu \mid \theta, \sigma^2 \sim N\left(\theta, \frac{\sigma^2}{n_0}\right).$$

Assume σ^2 and θ are known and fixed. The posterior distribution is:

$$p(\mu \mid y) \propto N\left(\mu \mid \theta, \frac{\sigma^2}{n_0}\right) \times \prod_{i=1}^n N(y_i \mid \mu, \sigma^2)$$
$$\propto N\left(\mu \mid \theta, \frac{\sigma^2}{n_0}\right) \times N\left(\bar{y} \mid \mu, \frac{\sigma^2}{n}\right)$$

A useful identity: Completing the square

$$a_1(x - b_1)^2 + a_2(x - b_2)^2 = A(x - B)^2 + C \text{ for all } x .$$

Find expressions for A , B and C in terms of a_1 , a_2 , b_1 and b_2 . The idea is simple.

- ▶ Expand the left-hand-side and the right-hand-side.
- ▶ Compare the coefficients for x^2 , x and the constant term (i.e., the coefficient of x^0).

$$(a_1 + a_2)x^2 - 2(a_1b_1 + a_2b_2)x + a_1b_1^2 + a_2b_2^2 = Ax^2 - 2ABx + AB^2 + C .$$

Comparing coefficients of x^2 we obtain

$$A = a_1 + a_2 .$$

Comparing coefficients of x we obtain $AB = a_1b_1 + a_2b_2$, which implies that

$$B = \frac{a_1b_1 + a_2b_2}{A} = \frac{a_1b_1 + a_2b_2}{a_1 + a_2} .$$

Comparing the constants we obtain $AB^2 + C = a_1b_1^2 + a_2b_2^2$. Therefore,

$$\begin{aligned} C &= a_1b_1^2 + a_2b_2^2 - AB^2 = a_1b_1^2 + a_2b_2^2 - (a_1 + a_2) \frac{(a_1b_1 + a_2b_2)^2}{(a_1 + a_2)^2} \\ &= a_1b_1^2 + a_2b_2^2 - \frac{(a_1b_1 + a_2b_2)^2}{a_1 + a_2} \end{aligned}$$

Expression for C : Further simplification.

$$\begin{aligned}C &= a_1 b_1^2 + a_2 b_2^2 - AB^2 = a_1 b_1^2 + a_2 b_2^2 - (a_1 + a_2) \frac{(a_1 b_1 + a_2 b_2)^2}{(a_1 + a_2)^2} \\&= a_1 b_1^2 + a_2 b_2^2 - \frac{(a_1 b_1 + a_2 b_2)^2}{a_1 + a_2} = \frac{(a_1 + a_2)(a_1 b_1^2 + a_2 b_2^2) - (a_1 b_1 + a_2 b_2)^2}{a_1 + a_2} \\&= \frac{\cancel{a_1^2 b_1^2} + a_1 a_2 b_2^2 + a_1 a_2 b_1^2 + \cancel{a_2^2 b_2^2} - \cancel{a_1^2 b_1^2} - \cancel{a_2^2 b_2^2} - 2a_1 a_2 b_1 b_2}{a_1 + a_2} \\&= \frac{a_1 a_2 b_2^2 + a_1 a_2 b_1^2 - 2a_1 a_2 b_1 b_2}{a_1 + a_2} = \frac{a_1 a_2 (b_2^2 + b_1^2 - 2b_1 b_2)}{a_1 + a_2} \\&= \frac{a_1 a_2}{a_1 + a_2} (b_2 - b_1)^2 .\end{aligned}$$

Completing the square identity:

$$\begin{aligned}a_1(x - b_1)^2 + a_2(x - b_2)^2 &= A(x - B)^2 + C \\&= (a_1 + a_2) \left(x - \frac{a_1 b_1 + a_2 b_2}{a_1 + a_2} \right)^2 + \frac{a_1 a_2}{a_1 + a_2} (b_2 - b_1)^2 .\end{aligned}$$

The above identity will be handy in subsequent calculations.

$$\begin{aligned}
p(\mu \mid y) &\propto N\left(\mu \mid \theta, \frac{\sigma^2}{n_0}\right) \times N\left(\bar{y} \mid \mu, \frac{\sigma^2}{n}\right) \\
&\propto \exp\left\{-\frac{n_0}{2\sigma^2}(\mu - \theta)^2\right\} \times \exp\left\{-\frac{n}{2\sigma^2}(\mu - \bar{y})^2\right\} \\
&\propto \exp\left\{-\frac{1}{2\sigma^2}\left[n_0(\mu - \theta)^2 + n(\mu - \bar{y})^2\right]\right\} \\
&\propto \exp\left\{-\frac{n_0 + n}{2\sigma^2}\left(\mu - \frac{n_0\theta + n\bar{y}}{n_0 + n}\right)^2\right\}
\end{aligned}$$

The left-hand side is a probability density (integrates to 1) and the right-hand side is proportional to a normal density in μ with mean $\frac{n_0\theta + n\bar{y}}{n_0 + n}$ and variance $\frac{\sigma^2}{n_0 + n}$. Hence,

$$p(\mu \mid y) = N\left(\mu \mid \frac{n_0\theta + n\bar{y}}{n_0 + n}, \frac{\sigma^2}{n_0 + n}\right)$$

- ▶ Consider the problem of estimating the current weight of a group of people. A sample of 10 people were taken and their average weight was calculated as $\bar{y} = 176$ lbs. Assume that the population standard deviation was known as $\sigma = 3$. Assuming that the data $y_1, \dots, y_{10}$ came from a $N(\mu, \sigma^2)$ population perform the following computations.
- ▶ Obtain a 95% confidence interval for μ using classical methods.
- ▶ Assume a prior distribution for μ of the form $N(\theta, \tau^2)$. Obtain 95% posterior credible intervals for μ for each of the cases: (a) $\theta = 176, \tau = 8$; (b) $\theta = 176, \tau = 1000$ (c) $\theta = 0, \tau = 1000$. Which case gives results closest to that obtained in the classical method?